

LLM-Assisted Intent-Aware Multimodal Video Retrieval with Diagnostic-Event Temporal Search

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Abstract. Large-scale video retrieval remains challenging due to the semantic gap between natural-language queries and visual content. In this report, we present our interactive video retrieval system developed for the Ho Chi Minh City AI Challenge 2026. The system integrates visual retrieval with textual evidence. To better adapt retrieval to query intent, an **LLM-based planner** generates semantic and lexical query views, decomposes complex queries into ordered events, and assigns query-dependent importance and modality weights. For temporal retrieval, we introduce **Diagnostic-Event-Video-First (DEV) search**, which identifies a discriminative event to reduce the candidate-video space before constructing temporally consistent event sequences under ordering and gap constraints. The system achieved RScore values of **11.6**, **13.4**, and **12.0** across the three preliminary rounds.

Keywords: Event retrieval · Temporal Search · LLM-Assisted Intent-Aware Multimodal Video Retrieval.

1 Introduction

Interactive video retrieval requires combining visual content, textual evidence, and temporal context to locate specific moments in large collections. Benchmarks such as the Video Browser Showdown, Lifelog Search Challenge, and Ho Chi Minh City AI Challenge (AIC) reflect these requirements [23, 24, 4]. AIthena-Vision and Vortex integrate multimodal retrieval with temporal search [15, 14]. Building on this foundation, we address two design questions: how to adapt evidence weights to query intent, and how to use distinctive events to guide multi-event sequence retrieval.

We present our interactive retrieval system developed for AIC 2026, supporting known-item search (KIS), question answering (QA), and Temporal Retrieval and Alignment of Key Events (TRAKE). The system combines OpenCLIP, an optional SigLIP2 branch, and metadata search over automatic speech recognition (ASR) transcripts, optical character recognition (OCR) outputs, and captions. A large language model (LLM) plans retrieval before accessing any retrieved evidence, while deterministic components perform ranking and temporal validation.

Our main contributions are:

- **Intent-aware multimodal planning.** A structured LLM planner generates semantic rewrites, lexical queries, event decompositions, and evidence weights for hybrid retrieval.
- **Diagnostic-event-guided temporal search.** Diagnostic-Event-Video-First (DEV-first) search combines planner priors with retrieval probes to prioritize candidate videos, followed by target-centered sequence recovery under chronological and temporal-gap constraints.

2 Related Work

Recent interactive video retrieval systems increasingly combine visual embeddings, textual metadata, and temporal context to support complex search scenarios. Althena-Vision integrates Perception Encoder features with OCR, ASR, object detection, LLM-based query expansion, and Adaptive Temporal Search for multi-event retrieval [15]. Its strength lies in diversifying query perspectives while exploiting multiple evidence sources. Vortex similarly combines CLIP and SigLIP2 through reciprocal rank fusion, together with OCR, captions, speech transcripts, relevance feedback, and temporal reranking [14]. VideoEase also adopts multimodal fusion using CLIP, BLIP2, and OpenCLIP, and supports temporal search between two scenes [25]. These systems show the effectiveness of multimodal evidence fusion, but modality importance is generally not explicitly determined from the semantic structure of each query, and multi-event temporal alignment remains only partially addressed.

Other approaches focus on richer event-level and contextual representations. U-CESE retrieves coherent clips using lightweight keyframe extraction, context-aware caption propagation, and a unified clipping strategy [13]. MEMORIA combines dense annotations, CLIP retrieval, event segmentation, and LLM-assisted query analysis to improve lifelog search [8], while LifeSearch incorporates multimodal metadata and structured spatial-temporal information [20]. SnapSeek 3.0 further introduces scene-graph-based retrieval and temporal consistency mechanisms [10]. T-EAGLE represents sequences of frames as textual action narratives, enabling retrieval at the event rather than isolated-frame level [16]. U-Cker complements these methods through interfaces that support interactive exploration of temporally related lifelog moments [27]. Although these approaches improve contextual and temporal retrieval, they mainly depend on predefined temporal modes, pre-generated event representations, or temporal grouping after retrieval.

Our system builds on these advances by connecting query understanding directly to evidence selection and temporal sequence construction. An LLM generates semantic rewrites, lexical queries, event decompositions, and modality-specific evidence weights before retrieval. Diagnostic-Event-Video-First search then uses event importance and preliminary retrieval evidence to identify promising videos, after which bounded beam search explicitly aligns candidates under chronological and temporal-gap constraints. The main distinction is therefore not any individual component, but the integration of query-dependent multimodal weighting, diagnostic-event-guided retrieval, and explicit multi-event temporal alignment within a unified pipeline.

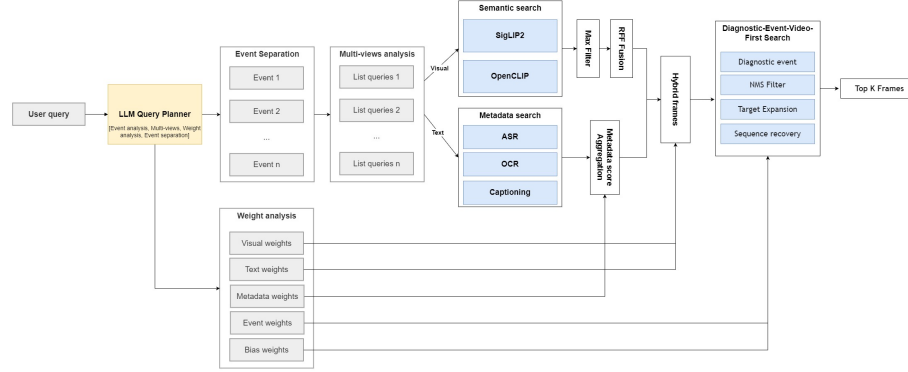


Fig. 1. Overview of the proposed multimodal video search pipeline.

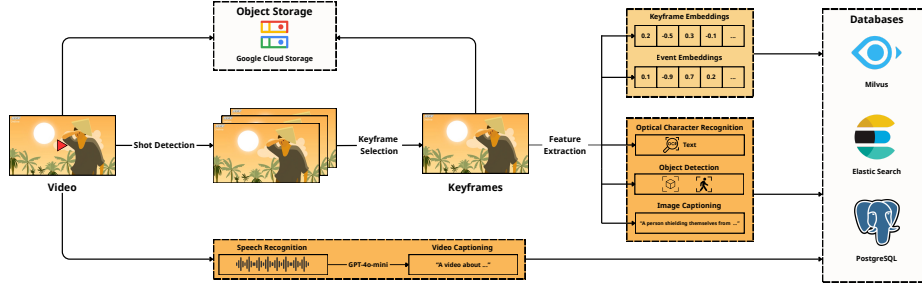


Fig. 2. Overview of the proposed data preprocessing pipeline.

3 Architecture

Figure 1 illustrates the proposed search pipeline with Coarse-to-fine Video-first Retrieval. Given a user query, the LLM planner decomposes it into ordered events, generates multiple semantic and lexical views, and assigns visual, metadata, and event-importance weights. Each event is searched through parallel vector-retrieval branches, while ASR, OCR, and caption metadata are retrieved by Elasticsearch. Visual results are combined by maximum-view filtering and reciprocal rank fusion when both encoders are active; metadata evidence is then merged with visual evidence to produce hybrid frame candidates. For temporal queries, Diagnostic-Event-Video-First (DEV) search selects a diagnostic event to identify candidate videos, removes redundant frames. Finally, timestamp constraints, sequence scoring, and diversification produce the top- K frames or video sequences.

3.1 Data Processing

Figure 2 illustrates our data processing pipeline. Given the raw videos, we first employ AutoShot [29] to identify shot boundaries. For each detected shot, three representative frames are sampled from the beginning, middle, and end portions. Compared with

using a single representative frame, this strategy provides broader temporal coverage and reduces the risk of missing transient but retrieval-relevant visual information. The extracted keyframes, together with the original videos, are stored in Google Cloud Storage [9] for efficient media access and visualization in the retrieval interface.

Visual representation. Each keyframe is independently encoded using OpenCLIP ViT-H/14-quickgelu with DFN-5B weights [11, 7] and SigLIP 2 SO400M/16 [26]. The DFN-5B checkpoint provides a strong CLIP-based image–text representation learned from carefully filtered large-scale image–text data, whereas SigLIP 2 improves semantic image–text retrieval and provides stronger multilingual and localized visual representations. Using these complementary embedding spaces increases the robustness of subsequent semantic retrieval. To obtain event-level representations, temporally adjacent keyframes whose embedding similarity exceeds a threshold T are grouped into the same event. Given an event containing N keyframes with embeddings $\{\mathbf{v}_i\}_{i=1}^N$, its representation is computed as

$$\mathbf{v}_e = \frac{1}{N} \sum_{i=1}^N \mathbf{v}_i. \quad (1)$$

This temporal aggregation reduces redundant frame-level representations while preserving the dominant semantics of each event for event-level retrieval.

Visual semantic enrichment. We further enrich each keyframe with complementary textual and object-level information. For optical character recognition (OCR), PaddleOCR [5] is employed for text detection, followed by VietOCR [19] for Vietnamese text recognition. The recognized text is subsequently normalized using the vietnamese-correction-v2 model [2] to mitigate spelling and recognition errors. Object information is extracted using the COCO-pretrained YOLO26x detector [12], which recognizes 80 common object categories. We adopt the largest YOLO26 detection variant to favor detection accuracy during offline preprocessing. In parallel, Qwen3-VL-4B-Instruct [1] generates keyframe captions that emphasize visible entities, their attributes, spatial relations, and interactions, providing richer semantic descriptions beyond isolated object labels.

Speech and video-level semantics. For the audio modality, speech segments are first identified using voice activity detection (VAD) and subsequently transcribed with Whisper Large V3 Turbo [22, 18]. VAD suppresses non-speech regions and provides finer speech segmentation, which is particularly useful for retaining short utterances. The resulting transcripts are post-processed by GPT-4o mini [17] for spelling normalization and English translation. In addition to local frame-level descriptions, we derive a global video-level semantic summary by providing the complete transcript of each video to GPT-4o mini and prompting it to extract its principal content. We refer to this transcript-derived summary as the *video caption*, which complements frame-level visual captions with global conversational context.

Indexing and storage. Finally, the resulting multimodal representations are organized according to their retrieval characteristics. Dense keyframe and embeddings are indexed

in Milvus [28] for vector similarity search, while textual metadata, including OCR outputs, captions, object labels, transcripts, and video-level summaries, are indexed in Elasticsearch [6] and maintained together with structured metadata in PostgreSQL [21]. This separation enables the retrieval system to efficiently combine visual similarity, textual matching, and structured filtering within a unified multimedia search pipeline.

3.2 Multimodal Retrieval

Semantic Vector Retrieval We use semantic vector retrieval to align natural-language queries with indexed video frames. For each keyframe I_f , we encode and ℓ_2 -normalize its visual representation as

$$\mathbf{v}_f^{(m)} = \frac{E_{\text{img}}^{(m)}(I_f)}{\|E_{\text{img}}^{(m)}(I_f)\|_2}, \quad (2)$$

where m denotes the embedding model. We compute these representations offline and store them in model-specific Milvus collections together with their video and keyframe identifiers.

We primarily use OpenCLIP ViT-H/14-quickgelu with DFN5B weights [7], producing 1024-dimensional embeddings, and additionally support SigLIP2 SO400M/16 with 1152-dimensional representations [26]. Since the two models define different embedding spaces, we maintain separate vector indexes for each model.

At retrieval time, we encode the original query and its LLM-generated semantic rewrites $\{q_j\}_{j=1}^J$ using the corresponding text encoder:

$$\mathbf{z}_j^{(m)} = \frac{E_{\text{text}}^{(m)}(q_j)}{\|E_{\text{text}}^{(m)}(q_j)\|_2}. \quad (3)$$

For each rewrite, we retrieve the top- k frames using cosine similarity. We then retain the strongest semantic evidence for each frame:

$$s_m(f) = \max_{\substack{1 \leq j \leq J \\ f \in \mathcal{N}_k^{(m)}(q_j)}} \left[\cos(\mathbf{z}_j^{(m)}, \mathbf{v}_f^{(m)}) \right]_+, \quad (4)$$

where $\mathcal{N}_k^{(m)}(q_j)$ denotes the top- k results for q_j , and $[\cdot]_+$ truncates negative similarities.

When multiple visual encoders are enabled, we combine their ranked results using weighted reciprocal rank fusion (RRF) [3]:

$$s_{\text{vec}}(f) = \sum_{m \in \mathcal{M}} \frac{\beta_m}{K + r_m(f)}, \quad (5)$$

where $r_m(f)$ and β_m denote the rank and weight associated with model m , respectively. With a single encoder, we directly use its semantic similarity score.

We further incorporate textual evidence retrieved from captions, OCR, and ASR. For the retained candidate set $\mathcal{C}$, we independently normalize the semantic and textual scores:

$$\hat{s}_x(f) = \frac{s_x(f)}{\max_{g \in \mathcal{C}} s_x(g)}, \quad x \in \{\text{vec}, \text{text}\}. \quad (6)$$

The hybrid ranker then combines these normalized scores with the configured modality weights and, when enabled, an additional reciprocal-rank fusion term.

Metadata Retrieval Visual retrieval may be insufficient for queries involving names, numbers, spoken content, or text appearing in the video. Therefore, we complement it with a metadata retrieval branch based on Elasticsearch. For each keyframe, metadata is extracted from automatic speech recognition (ASR), optical character recognition (OCR), and image captioning.

The query planner generates lexical variants for ASR/OCR and English semantic rewrites for caption retrieval. Elasticsearch performs boosted search over these sources. For a keyframe f , we define the metadata score as

$$S_{\text{meta}}(f, q) = \max_{\substack{s \in \mathcal{S} \\ q_i \in \mathcal{Q}_s}} [b_s w_s S_{\text{ES}}(f, q_i, s)], \quad (7)$$

where $\mathcal{S} = \{\text{asr}, \text{ocr}, \text{caption}\}$, S_{ES} denotes the Elasticsearch relevance score, and b_s and w_s are the source-specific boost and adaptive weights, respectively.

We then combine metadata and visual evidence as

$$S_{\text{hyb}}(f, q) = \alpha S_{\text{vis}}(f, q) + \beta S_{\text{meta}}(f, q) + \gamma S_{\text{qual}}(f). \quad (8)$$

Since the two retrieval branches operate on different score scales, we additionally apply Reciprocal Rank Fusion (RRF):

$$S_{\text{RRF}}(f) = \frac{\alpha}{k + r_{\text{vis}}(f)} + \frac{\beta}{k + r_{\text{meta}}(f)}, \quad (9)$$

where $k = 60$. This hybrid formulation supports both visual-semantic queries and queries that rely on textual evidence such as dialogue, signs, numbers, or named entities.

3.3 LLM-Assisted Retrieval Planning

We employ LLM as a query planner to transform user descriptions into structured retrieval representations. Given a query and its task type, the LLM generates plan including semantic rewrites, lexical queries, and modality weights. Semantic rewrites are expressed in English view for vision-language retrieval, while integrating metadata evidences, such as objects, actions, attributes (colors, counts, shapes, sizes) to ensure alignment evidences in metadata to better match OCR and ASR annotations.

For KIS and QA, we construct a query-level retrieval plan. For TRAKE and temporal KIS, we decompose the query into an ordered set of M events: $\mathcal{E} = \{e_1, e_2, \dots, e_M\}$, where each event contains complementary retrieval views, an importance score, diagnostic prior coefficients and modality evidence weights.

System workflow. The planning procedure consists of four stages: temporal decomposition, structured query interpretation, evidence-aware retrieval routing, event-aware reasoning.

Stage 1: Temporal and Bilingual Multi-View Decomposition. For temporal KIS and TRAKE queries, the planner decomposes the description into chronologically ordered, independently retrievable events $\mathcal{E} = (e_1, e_2, \dots, e_N)$. Each event we use LLM to generates k views with Chain-Of-Thought (CoT) rules ensuring rich in information (subject, actions, object and evidence factors (color, scene, attributes)).

Stage 2: Structured Query Interpretation. Given an input query q and its task type $\tau \in \{\text{KIS, QA, TRAKE}\}$, the LLM planner converts the description into a structured retrieval plan: $\mathcal{P}_i = (q_{e_i}, \mathcal{V}_i^{\text{en}}, \mathcal{V}_i^{\text{vi}}, \mathcal{T}_i, I_i, D_i, \mathbf{w}_i^{\text{modality}}, \mathbf{w}_i^{\text{metadata}})$, where q_{e_i} is the standalone event query, $\mathcal{V}_i^{\text{en}}$ denotes English semantic views, $\mathcal{V}_i^{\text{vi}}$ denotes Vietnamese semantic views, and $\mathcal{T}_i$ contains lexical queries used by text-based retrieval. I_i is the event *importance* measures the contribution of an event to the requested final result, whereas D_i is its *diagnostic prior* which estimates how useful the event is for identifying the correct video during the coarse retrieval stage. Set of modality weights $\mathbf{w}_i^{\text{modality}}$ for visual and text retrieval scores and set of metadata weights $\mathbf{w}_i^{\text{metadata}}$ for OCR, Caption and ASR retrieval scores.

Stage 3: Evidence-Aware Retrieval Routing. The planner distinguishes between visual and textual evidence rather than applying a fixed multimodal weighting to every query. At both the query level and the event level, it predicts $\mathbf{w}_i = (w_i^{\text{vis}}, w_i^{\text{text}})$, where w_i^{vis} controls visual-semantic retrieval and w_i^{text} controls metadata-based retrieval ($w_i^{\text{vis}} + w_i^{\text{text}} = 1$). Textual evidence is further decomposed into ASR, caption, and OCR sources: $\mathbf{w}_i^{\text{metadata}} = (w_i^{\text{ASR}}, w_i^{\text{CAP}}, w_i^{\text{OCR}})$, where $w_i^{\text{ASR}} + w_i^{\text{CAP}} + w_i^{\text{OCR}} = 1$.

Stage 4: Event-Aware Reasoning. Temporal structure is represented explicitly by adjacent event edges $g_i = (e_i, e_{i+1}, r_i, c_i)$, where r_i denotes the temporal relation and $c_i \in \{c_{\text{short}}, c_{\text{medium}}, c_{\text{long}}, c_{\text{unknown}}\}$ represents a semantic gap class to get information about relative temporal constraints without hallucinating exact time durations. The plan additionally distinguishes between a complete video sequence and a specific target frame through temporal intent, anchor policy.

3.4 Temporal Search

In AI Challenge, to solve the lack of videos in top K when searching, we introduces **Diagnostic Event Video-first search (DEV)** for KIS and TRAKE retrieval problems with a **Coarse-to-Fine** temporal retrieval strategy. It estimates firstly which events are most useful for identifying the source video, selects a diverse candidate-video set from global evidence (Global search), and then performs a event-level retrieval (Local search) restricted to those videos.

For each event e_i , the planner provides an importance value w_i , a diagnostic prior p_i , event-specific retrieval parameters η_i , and semantic query views.

We define the retrieval operator as $R(\mathcal{P}_i; b, \mathcal{V})$, where b is the candidate budget and $\mathcal{V}$ is an optional set of allowed videos. Global retrieval is obtained when $\mathcal{V} = \mathcal{V}_{\text{all}}$, whereas DEV's second-stage retrieval uses a restricted candidate-video set $\mathcal{V}^*$.

Stage 1: Global event probing. DEV first retrieves a global candidate set for every planned event, $C_i^g = R(\mathcal{P}_i; b_g, \mathcal{V}_{\text{all}})$, where $i = 1, \dots, M$. DEV calibrates each candidate using its score based on relative rank $r_i(f)$ (rank of frame f for event e_i) and scores from multimodal retrieval with heuristic coefficients.

Stage 2: Diagnostic event and candidate-video selection. A lightweight retrieval probe estimates how discriminative each event is. For the top candidates of e_i , let N_i be the number of candidates and $\mathcal{U}_i$ the set of distinct retrieved videos. DEV computes the video-selectivity signal $u_i = \text{clip}_{[0,1]} \left(1 - \frac{|\mathcal{U}_i|}{N_i} \right)$. Let $s_{i,(1)} \geq s_{i,(2)} \geq \dots$ be the calibrated scores sorted in descending order. The top-confidence signal is $c_i^{\text{conf}} = s_{i,(1)}$, and the score-margin signal m_i . The deterministic probe score is $\phi_i = h_u u_i + h_m m_i + h_c c_i^{\text{conf}}$. DEV combines this retrieval-time evidence with the LLM-generated diagnostic prior D_i to compute score: $\rho_i = h_D D_i + h_\phi \phi_i$. We retain the two highest-scoring diagnostic events. To reduce redundant evidence, temporal non-maximum suppression is applied within each (v, e_i) group, followed by aggregating the global evidence at video level. Then, we compute the score of each video: $S_{\text{video}}(v) = w_A A_v + w_X X_v + w_C C_v$. Then, DEV filters videos based on scores and save to set of videos $\mathcal{V}^*$.

Stage 3: Video-local retrieval. Second restricted video retrieval rather than a post filtering operation over the global search results. The local NMS window and retains a computed candidates for each (v, e_i) group. Starting from the primary diagnostic event, DEV greedily expands to preceding and subsequent events while enforcing chronological order and LLM-provided temporal gap constraints. For timestamped candidates, a frame f is retained only when $t(f_j) - t(f_i) \leq \Delta(e_i)$ where the planner gap classes $c_i \in C_{\text{timestamp}}$.

Stage 4: Sequence scoring. For each diagnostic candidate in a video, DEV greedily expands the sequence in both temporal directions with anchor frame. Each recovered sequence π is scored using weighted event evidence, event coverage, then the resulting sequences are diversified and ranked to produce the final top- K results.

Pseudocode

Algorithm 1 Diagnostic Event Video-first Search (DEV)

Require: Ordered events $\mathcal{E} = (e_1, \dots, e_M)$, event plans $\{\mathcal{P}_i\}_{i=1}^M$, importance $\{I_i\}$, diagnostic priors $\{D_i\}$, temporal relation G , output size K , Set of heuristic configures $\mathcal{H}$

Ensure: Ranked temporal sequences Π

```

1:  $\Pi \leftarrow \emptyset$ 
2: Stage 1: Global event probing
3: for  $i = 1$  to  $M$  do
4:    $C_i^g \leftarrow R(\mathcal{P}_i; b_g, \mathcal{V}_{\text{all}})$ ;  $C_i^g \leftarrow \text{CALIBRATECANDIDATES}(C_i^g)$ 
5:    $(u_i, m_i, c_i^{\text{conf}}) \leftarrow \text{EVENTPROBE}(C_i^g)$ 
6:    $\phi_i \leftarrow h_u u_i + h_m m_i + h_c c_i^{\text{conf}}$ ;  $\rho_i \leftarrow h_D D_i + h_\phi \phi_i$ 
7: end for
8:  $\mathcal{D} \leftarrow \text{TOPDIAGNOSTICEVENTS}(\{\rho_i\})$ 
9: for all  $i \in \mathcal{D}$  do
10:   $C_i^g \leftarrow \text{TEMPORALNMS}(C_i^g)$ 
11: end for

12: Stage 2: Candidate-video selection
13: Normalize  $\{I_i\}$  to obtain  $\{\bar{I}_i\}$ 
14: for all video  $v$  occurring in the global candidates do
15:   for  $i = 1$  to  $M$  do
16:      $x_{v,i} \leftarrow \max_{f \in C_i^g: v(f)=v} s_i(f)$ 
17:   end for
18:    $A_v \leftarrow x_{v,d}$ ;  $X_v \leftarrow \sum_i \bar{I}_i x_{v,i}$ ;  $C_v \leftarrow \sum_i \bar{I}_i \mathbf{1}[x_{v,i} > 0]$ 
19:    $S_{\text{video}}(v) \leftarrow w_A A_v + w_X X_v + w_C C_v$ 
20: end for
21:  $\mathcal{V}^* \leftarrow \text{DIVERSEVIDEOSELECTION}(\{C_i^g\}, S_{\text{video}}, \{I_i\})$ 

22: Stage 3: Video-local retrieval
23: for  $i = 1$  to  $M$  do
24:    $C_i^l \leftarrow R(\mathcal{P}_i; b_l, \mathcal{V}^*)$ ;  $C_i^l \leftarrow \text{CALIBRATECANDIDATES}(C_i^l)$ 
25:    $C_i^l \leftarrow \text{TEMPORALNMS}(C_i^l)$ 
26: end for
27:  $m_{\min} \leftarrow \text{ADAPTIVEMINMATCH}(\{C_i^l\}, \{D_i\})$ 

28: Stage 4: Sequence scoring
29:  $\Pi \leftarrow \text{RECOVERSEQUENCES}(\mathcal{V}^*, \{C_i^l\}_{i=1}^M, d, G, m_{\min})$ 
30:  $\Pi \leftarrow \text{TWOTIERRANK}(\Pi, K)$ 
31: return top- $K$  sequences from  $\Pi$ 

```

3.5 User Interface

Our user interface is designed to support efficient human-in-the-loop video retrieval, as illustrated in Fig. 3. The interface is organized into four main regions. **(A) Task Navigation** provides access to KIS, QA, TRAKE, and video retrieval modes. **(B) Retrieval Workspace** presents ranked keyframes together with their retrieval scores and allows users to inspect and select candidate frames. **(C) Query Panel** contains the textual query input and search controls, enabling users to formulate or refine the retrieval

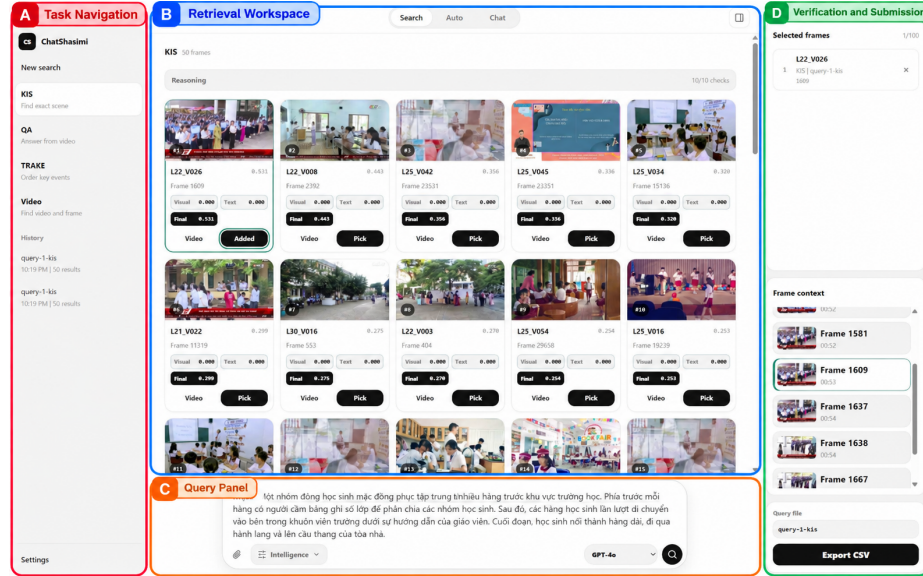


Fig. 3. Overview of the proposed video retrieval interface. (A) task navigation, (B) ranked retrieval results, (C) query input and search controls, and (D) frame verification, temporal context, and result export.

request. (D) **Verification and Submission** displays selected frames and their temporal context, enabling final verification before exporting the retrieval results.

4 Experiments

Research question.

- RQ1: What are the effectiveness and latency trade-offs of retrieving videos first and refining frame details afterwards, compared with direct temporal frame retrieval?
- RQ2: How do the choice of semantic embedding model and the number of LLM-generated query perspectives affect retrieval effectiveness and latency?
- RQ3: How sensitive is retrieval performance to individual components of the LLM retrieval-planner prompt?

4.1 Temporal search algorithms (RQ1)

Experimental setup We summarize and label AIC 2026 ground-truth benchmark with 78 queries, comprising 58 KIS temporal queries and 20 QA queries to make dataset for this experiments.⁴ We evaluated with Recall@ k for $k \in \{1, 5, 10, 20, 50, 100\}$ and median latency. We compare 3 algorithm ATS, Vortex, and DEV, where DEV performs

⁴ Recall@ k measures whether the result is retrieved within the first k candidates.

global retrieval and video selection before local frame refinement. All runs use the recorded fixed configuration: GPT-4o planning, SigLIP2 visual search, temporal mode enabled, top-100 retrieval.

Table 1. Recall@k and latency comparison for temporal search strategies.

Method	R@1	R@5	R@10	R@20	R@50	R@100	Latency (ms)
ATS	41.03	64.10	67.95	73.08	78.21	80.77	13320.0
Vortex	37.18	61.54	69.23	75.64	82.05	88.46	10832.5
DEV	34.62	60.26	76.92	82.05	85.90	93.59	37410.0

Result Table 1 shows that ATS performs best at early ranks, achieving the highest R@1 and R@5. In contrast, DEV obtains the best recall from R@10 onward, reaching 93.59% at R@100, compared with 88.46% for Vortex and 80.77% for ATS. This indicates that DEV preserves more relevant candidates at larger retrieval depths.

However, this improvement comes with higher computational cost. DEV has a median latency of 37,410 ms, compared with 13,320 ms for ATS and 10,832.5 ms for Vortex, showing a trade-off between retrieval recall and runtime efficiency.

4.2 Semantic embedding models (RQ2)

Experimental setup We evaluate the effect of the visual encoder and LLM-generated multi-view query expansion on semantic retrieval. The ablation is conducted on 58 queries using the two visual encoders supported by our system, OpenCLIP and SigLIP2. For each encoder, we compare the original full query with perspective-based retrieval using $n \in \{3, 5, 7\}$ semantic views.⁵ We additionally report mean retrieval latency to quantify the computational cost of increasing the number of perspectives.

Table 2. Visual encoder choice and multi-perspective query expansion over 58 queries.

Encoder	Mode	n	MRR $\uparrow$	R@10 $\uparrow$	R@100 $\uparrow$	Lat. (ms) $\downarrow$
OpenCLIP	Full	1	0.0382	0.0862	0.1034	<u>150.1</u>
	View.	3	0.0067	0.0172	0.0690	207.7
	View.	5	0.0062	0.0172	0.0517	229.7
	View.	7	0.0062	0.0172	0.0517	247.6
SigLIP2	Full	1	<u>0.0617</u>	<u>0.1034</u>	<u>0.1379</u>	158.3
	View.	3	0.0371	<u>0.1034</u>	<u>0.1379</u>	210.7
	View.	5	0.0371	<u>0.1034</u>	<u>0.1379</u>	230.8
	View.	7	0.0369	0.0862	<u>0.1379</u>	249.7

⁵ Retrieval effectiveness is measured using mean reciprocal rank (MRR), Recall@10 (R@10), and Recall@100 (R@100).

Result As shown in Table 2, full-query retrieval outperforms multi-perspective expansion for OpenCLIP, while also achieving lower latency. SigLIP2 is more robust: three or five perspectives preserve R@10 and R@100, although MRR decreases. Overall, SigLIP2 with the full query achieves the strongest retrieval effectiveness, whereas OpenCLIP with the full query provides the lowest latency. Increasing the number of perspectives beyond three yields no additional recall gains and consistently increases latency.

4.3 Prompt versions for LLM Retrieval Planner (RQ2)

We hold the retrieval stack fixed and ablate the planner prompt. The complete repository prompt is the *Full Prompt*; each variant removes one configured component C_i . The components are C1 meaning preservation, C2 clause classification, C3 global weights, C4 text routing, C5 modality preference, C6 English semantic views, C7 Vietnamese lexical views, C8 OCR preservation, C9 observable terms, C10 rewrite length, C11 root rewrite, C12 temporal-KIS planning, C13 task-specific event structure (including ordered-query and TRAKE constraints), and C14 event weighting.

We use `gpt-4o-mini-2024-07-18`, temperature 0.05, and a 2,048-token output limit. Three repetitions over 15 conditions yield $15 \times 3 \times 98 = 4,410$ successful canonical records. All other retrieval settings are fixed (SigLIP2, strict hybrid top-100, metadata enabled, temporal mode for KIS, and reranking and query expansion disabled).

For each query, video-level Recall@ k indicates whether the first k results contain the ground-truth video. The primary outcome is the unweighted mean of Recall@1, @5, @20, @50, and @100 (MeanR). Exact-frame recall, using the single annotated frame, is secondary and is averaged over the same cutoffs. We define $\Delta = \text{ablation} - \text{Full}$. Repetitions are first averaged within query and condition, after which Full and each ablation are paired by query.

We use 10,000 paired bootstrap samples for the 95% CI and 20,000 paired random sign flips for the two-sided p -value.

The Full Prompt obtains Recall@1, @5, @20, @50, and @100 of 35.71%, 51.70%, 70.41%, 73.81%, and 77.21%, respectively, giving a MeanR of 61.77% and mean exact-frame recall of 5.71%. Table 3 reports all 14 removals.

Table 3. Leave-one-component-out planner-prompt ablation. MeanR is in percent; Δ and 95% CIs are percentage points relative to Full.

ID	MeanR	Δ	95% CI	p	ID	MeanR	Δ	95% CI	p
C1	62.31	+0.54	[−2.38, 3.54]	.755	C8	63.27	+1.50	[−2.04, 5.10]	.449
C2	63.95	+2.18	[−1.70, 6.12]	.292	C9	60.75	−1.02	[−5.10, 2.79]	.646
C3	63.74	+1.97	[−1.36, 5.37]	.276	C10	63.74	+1.97	[−1.77, 6.05]	.344
C4	63.33	+1.56	[−2.38, 5.65]	.473	C11	64.42	+2.65	[−0.75, 6.26]	.160
C5	62.72	+0.95	[−2.38, 4.35]	.611	C12	63.88	+2.11	[−2.11, 6.19]	.339
C6	63.54	+1.77	[−3.27, 7.01]	.527	C13	63.40	+1.63	[−1.97, 5.31]	.415
C7	60.34	−1.43	[−5.03, 2.04]	.466	C14	62.99	+1.22	[−3.27, 5.78]	.626

All CIs include zero and all sign-flip tests have $p > .05$; hence, no component effect is statistically conclusive at the 5% level. Twelve removals have positive MeanR point estimates, whereas C7 and C9 are negative. C7 gives the largest directional decrease (-1.43 points; -2.58 within KIS), while C9 reduces MeanR by 1.02 points and mean exact-frame recall by 1.84 points. These subgroup patterns are descriptive, especially for the 23-query QA subset, and measure conditional sensitivity within the complete prompt rather than isolated causal importance.

MeanR can hide ranking-depth trade-offs: removing C3 increases MeanR by 1.97 points and Recall@5 by 6.46 points but decreases Recall@1 by 3.74 points; removing C12 increases MeanR by 2.11 points and Recall@100 by 6.12 points while decreasing Recall@1 by 3.40 points and mean exact-frame recall by 3.13 points.

Thus, a positive delta does not imply improvement at every rank or that a component is unnecessary, particularly because prompt components may interact.

The 14 related tests are unadjusted for multiplicity, but all unadjusted p -values already exceed .05, so correction would not change the non-significance conclusion. No TRAKE-effectiveness claim can be made from this benchmark; the small QA subset and single-reference exact-frame metric further limit generalization.

4.4 Analysis

Table 4 summarizes the official preliminary-round results of our system in the AIC 2026 Multimodal Retrieval Challenge. The system achieved scores of 21.27, 27.20, and 27.75 across the three evaluation rounds, corresponding to 88.63%, 93.79%, and 84.09% of the maximum scores, respectively. Overall, the system obtained 76.22 out of 86 points, equivalent to 88.63% of the total available score. The results demonstrate the effectiveness of the proposed multimodal and temporal retrieval pipeline across multiple evaluation rounds. The lower relative performance in Round 3 indicates that further improvements are still possible for more complex retrieval scenarios.

Table 4. Official preliminary-round results on the AIC 2026 benchmark.

Round	Official Score	Maximum Score	Score (%)
Round 1	21.27	24	88.63%
Round 2	27.20	29	93.79%
Round 3	27.75	33	84.09%
Overall	76.22	86	88.63%

5 Conclusion

We proposed an interactive multimodal video retrieval system for AI Challenge 2026 that combines LLM-assisted query planning, hybrid visual-textual retrieval, and DEV-first temporal search. The system achieved stable performance across three preliminary rounds and effectively narrowed candidate videos for complex KIS and TRAKE queries. Remaining challenges include camera transitions and visually ambiguous temporal sequences, motivating more adaptive retrieval strategies in future work.

5.1 Limits

5.2 Future Work

A natural extension of the current system is to move from one-shot LLM-assisted planning toward an agentic retrieval framework. The current planner determines query reformulations, evidence preferences, and event decompositions before retrieval, while difficult cases may still require manual inspection and query reformulation. Future work will investigate an iterative retrieval agent that can observe intermediate evidence, select among visual, textual, and temporal retrieval tools, and refine its retrieval strategy accordingly. Such a closed-loop formulation may reduce manual intervention and improve robustness for ambiguous or multi-event queries, while retaining deterministic retrieval and temporal validation where appropriate.

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